

Kidist Amde Mekonnen

PhD Researcher in Information Retrieval, Recommendation & Large-Scale Machine Learning

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 Amsterdam, NL

SUMMARY

PhD researcher at the **University of Amsterdam IRLab** working on generative retrieval, recommendation, neural information retrieval, and large-scale machine learning. Published at **SIGIR 2025**, **SIGIR 2026**, and **ACL Findings 2025**. Experienced in building end-to-end research systems with **PyTorch**, **FAISS**, **Hugging Face**, **Docker**, and **Slurm**. Research interests include retrieval, recommender systems, LLMs, multilingual NLP, and scalable representation learning.

RESEARCH EXPERIENCE

PhD Researcher

Oct 2023 – Present

University of Amsterdam, IRLab

Amsterdam, NL

- Developed methods for **continual generative retrieval** over dynamic corpora, mitigating catastrophic forgetting during sequential index/model updates; accepted as a **SIGIR 2026 full paper**.
- Reproduced and stress-tested **look-ahead decoding** for generative retrieval under query variation, typographical noise, and cross-lingual shift; accepted in the **SIGIR 2026 Reproducibility Track**.
- Designed **DDRO**, a lightweight pairwise relevance optimization objective for generative retrieval, improving **MRR@10 by 7.4% on MS MARCO** and **19.9% on Natural Questions**.
- Led Amharic passage retrieval benchmarking, including dataset construction, model training, and evaluation; improved dense retrieval **MRR@10 by 5.01%** over strong multilingual baselines.
- Built scalable training and evaluation pipelines for retrieval experiments, including ablations, error analysis, re-ranking, and reproducibility checks.

Data Science Research Intern

Apr 2022 – Oct 2022

Nanoverly Limited

Newcastle upon Tyne, UK

- Implemented and evaluated **GAN-based** models for synthetic DNA strand generation for nanorobot probe design.
- Developed predictive models for forecasting sensor behavior across laboratory environments and communicated results to research stakeholders.

TECHNICAL SKILLS

- Languages:** Python, Java, C++, R
- ML/IR:** PyTorch, TensorFlow, Hugging Face, FAISS, Elasticsearch, dense retrieval, sparse retrieval, learning-to-rank, recommender systems, LLMs
- Systems:** Docker, Slurm, Git, Linux/Unix, large-scale experimentation
- Methods:** ablation studies, error analysis, benchmarking, reproducible research

EDUCATION

PhD Candidate, Information Retrieval

Oct 2023 – Present

University of Amsterdam

Amsterdam, NL

MSc in Data Science

2020 – 2023

University of Trento

Trento, Italy

MSc in Machine Intelligence

2019 – 2020

AIMS-AMMI

Kigali, Rwanda

BSc in Computer Science

2013 – 2017

University of Gondar

Gondar, Ethiopia

SELECTED PUBLICATIONS

- Mekonnen, K.A., Tang, Y., and de Rijke, M. *A Parametric Memory Head for Continual Generative Retrieval*. **SIGIR 2026, Full Paper Track**. doi
- Mekonnen, K.A., Li, Y., Tang, Y., Lupart, S., and de Rijke, M. *Lost in Decoding? Reproducing and Stress-Testing the Look-Ahead Prior in Generative Retrieval*. **SIGIR 2026, Reproducibility Track**. doi
- Mekonnen, K.A., Tang, Y., and de Rijke, M. *Lightweight and Direct Document Relevance Optimization for Generative Information Retrieval*. **SIGIR 2025, Full Paper Track**. doi
- Mekonnen, K.A., Alemneh, E., and de Rijke, M. *Optimized Text Embeddings & Benchmarks for Amharic Passage Retrieval*. **ACL Findings 2025**. link

SELECTED PROJECTS

Semantic Identifier Search for Recommendation

2025 –

Present

Research Project

- Investigated semantic IDs, trie-constrained generation, and centroid-guided beam search for sequential recommendation.
- Built FAISS-based retrieval and evaluation pipelines for large-scale recommendation experiments.

Spatiotemporal Mobility Analytics

2023

MSc Project

- Built a spatiotemporal analytics pipeline for GPS and time-diary data, including trajectory cleaning, temporal aggregation, subgroup analysis, and interactive visualization.  Code

Birth-rate Estimation Pipeline

2023

MSc Project

- Developed an ML pipeline using Twitter and Eurostat data, including data collection, Flask-based labeling, BERT classification, and regression-based birth-rate estimation.  Code

TEACHING & SERVICE

Teaching Assistant

2024 – Present

University of Amsterdam

- TA for **Information Retrieval and Recommender Systems**; supported labs, assignments, projects, and exam grading.

Assistant Lecturer

2017 – 2019

University of Gondar

- Lectured computer science courses and supervised labs, assignments, and undergraduate projects.

Academic Service & Mentoring

2019 – Present

- Reviewer for **SIGIR** and **ECIR**; received **ECIR best reviewer recognition**.
- Mentor: **ELLIS ESSIR'24** , **AddisCoder** , and **MeetMentors.org** .

SELECTED ADDITIONAL WORK

- Lupart, S., Mekonnen, K.A., Abbasianteab, Z., and Aliannejadi, M. *uvav-irlab-conv at SemEval-2026 Task 8: Multi-Turn RAG with Learned Sparse Retrieval and Listwise Reranking*. **SemEval 2026**.
- Alemneh, Y.W., Mekonnen, K.A., and de Rijke, M. *The Multilingual Curse at the Retrieval Layer: Evidence from Amharic*. **MeLLM @ ACL 2026**.
- Mekonnen, K.A. *Adv-KD: Adversarial Knowledge Distillation for Faster Diffusion Sampling*. **arXiv**.
- Mekonnen, K.A. *Balanced Face Dataset: Guiding StyleGAN for Labeled Synthetic Face Image Dataset for Underrepresented Group*. **arXiv**.